

Yan Ma

600+ citations, 1.3k+ GitHub stars across research projects

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RESEARCH INTERESTS

My research aims to build multimodal agents for partially observed environments, especially how they perceive, reason, act, and predict over long-horizon tasks. So far, my work has mainly studied how RL changes multimodal models, including training dynamics, the joint improvement of reasoning and perception, and agentic vision tool use. Recently, I am interested in predictive multimodal learning from video, especially how video data can be cleaned, filtered, and annotated to better support the anticipation of future observations and action outcomes.

EDUCATION

- **Ph.D. Student, Fudan University** (Advisors: Prof. Pengfei Liu and Prof. Yu Qiao) 2023.09 – 2027.06
- **Master in Computer Application Technology, Fudan University** 2020.09 – 2023.06
- **Bachelor in Computer Science, Dalian University of Technology** (GPA: 4.076/5) 2016.09 – 2020.06

EXPERIENCE

- **Shanghai Innovation Institute** (2026.03 – 2026.06): leading a large-scale audio-video data processing pipeline
- **MiniMax & HailuoAI** (2025.01 – 2026.02): worked on multimodal foundation models for visual reasoning and vision tool use in the M-series, and image caption systems for Hailuo video generation
- **Shanghai AI Laboratory** (2023.07 – 2025.01): worked on text generation and unified multimodal model training

PUBLICATIONS

- **What Does Vision Tool-Use Reinforcement Learning Really Learn? Disentangling Tool-Induced and Intrinsic Effects for Crop-and-Zoom** [A][C]. *ICML 2026*
Yan Ma[†], Weiyu Zhang[†], Tianle Li, Linge Du, Xuyang Shen, Pengfei Liu
TLDR: We show current vision tool-use RL gains are driven mainly by intrinsic capability improvement and reduced tool-induced harm, not true tool-based correction.
- **One RL to See Them All: Visual Triple Unified Reinforcement Learning** [A][C]. *arXiv:2505.18129* [COLM 2026 under review]
Yan Ma[†], Linge Du[†], Xuyang Shen[†], Shaoxiang Chen, Pengfei Li, Qibing Ren, Lizhuang Ma, Yuchao Dai, Pengfei Liu, Junjie Yan
TLDR: We build a unified RL system (V-Triune) for jointly training visual reasoning and perception in VLMs, yielding broad gains across both task types.
- **Rethinking RL Scaling for Vision Language Models: A Transparent, From-Scratch Framework and Comprehensive Evaluation Scheme** [A][C]. *arXiv:2504.02587*
Yan Ma, Steffi Chern, Xuyang Shen, Yiran Zhong, Pengfei Liu
TLDR: We introduce a transparent, from-scratch RL framework and trajectory-level evaluation for VLMs, showing stronger generalization than SFT and clearer training-dynamics insights.
- **ANOLE: An Open, Autoregressive, Native Large Multimodal Models for Interleaved Image-Text Generation** [A][C][C]. *arXiv:2407.06135; extended version accepted to ICLR 2025 Blog Track*
Ethan Chern*, Jiadi Su*, Yan Ma*, Pengfei Liu
TLDR: We present an open native autoregressive multimodal model for coherent interleaved image-text generation, with efficient fine-tuning from Chameleon.
- **MoPS: Modular Story Premise Synthesis for Open-Ended Automatic Story Generation** [A][C]. *ACL 2024*
Yan Ma, Yu Qiao, Pengfei Liu
- **Weak-to-Strong Reasoning** [A][C]. *EMNLP Findings 2024*
Yuqing Yang, Yan Ma, Pengfei Liu

TECHNICAL SKILLS & ACADEMIC SERVICE

- Verl, Transformers, Ray, FSDP, vLLM, SGLang, Jax, FFmpeg, Codex, Git, NeoVim, Tmux
- Reviewer for AAAI 2025, NeurIPS 2025, COLM 2025–2026, ICLR 2026, ICML 2026, and ACL ARR 2026